

Cole Schreiner

Portfolio: [UserIsMissing.GitHub.io](https://UserIsMissing.github.io)

[in](https://www.linkedin.com/in/cole-schreiner) [Linkedin.com/in/cole-schreiner](https://www.linkedin.com/in/cole-schreiner) [✉](mailto:ColeRobotEngineer@gmail.com) ColeRobotEngineer@gmail.com

Professional Profile

Robotics and electrical engineering specialist completing a Master's at UC Santa Cruz with hands-on experience building embedded systems, robotic platforms, research instruments, and wearable devices from concept through validation.

Project background spans autonomous robotics, environmental control systems, sensor fusion, microcontroller programming, real-time control, and CAD-driven mechanical design.

Experienced leading small interdisciplinary teams, mentoring technical work across disciplines, defining subsystem interfaces, and delivering complete prototypes under academic and competition deadlines.

Education

University of California, Santa Cruz

Expected June 2026

Robotics Concentration ~ Graduated with Honors

- Master of Science in Electrical and Computer Engineering
- Bachelor of Science in Robotics Engineering
- Minor in Electrical Engineering
- Key Coursework: Robot kinematics and dynamics, feedback control systems, microcontroller system design, VLSI, circuit theory, embedded programming, and machine learning for engineered systems.

Technical Skills

Languages: C, C++, Java, Python, Verilog, MATLAB

Embedded and Control: ESP32, Raspberry Pi, PIC32, Arduino, FreeRTOS, PID control, finite state machines, real-time sensing, data logging

Design and Fabrication: SolidWorks, OnShape, AutoDesk, 3D printing, laser cutting, prototyping, electronics assembly, instrumentation and oscilloscope-based debugging

Software and Analysis: Git, Vivado, LTSpice, PyTorch, GUI development, web-based interfaces, signal processing, peak detection, normalization, and model training

Circuit and Systems: Analog filters, op-amps, power regulation, motor drivers, feedback control, RL/RLC analysis, frequency response, and sensor signal conditioning

Engineering Areas: Robotics, mechatronics, kinematics, microcontroller system design, VLSI, scientific instrumentation, machine learning, and imitation learning

Selected Engineering Projects

UAV Artificial Intelligence Autopilot and Simulation [\[Project Link\]](#)

February, 2026

Neural Flight Control and Behavioral Cloning

- Reframed a traditional small-scale UAV autopilot problem by replacing successive loop-closure control logic with a recurrent neural network trained through behavioral cloning.
- Built a PyTorch inference pipeline that mapped 15 flight-state inputs, including attitude, body-frame velocity, and airflow features, to continuous throttle, aileron, elevator, and rudder commands.
- Integrated the learned controller into a university flight-simulation framework to evaluate autonomous path following, closed-loop stability, and disturbance rejection under simulated wind conditions.
- Applied data normalization and mean-squared-error training against expert teacher demonstrations to validate whether a single learned controller could manage coupled flight dynamics in real time.

Capstone - Wet-Dry Cycler (RNA Replicator) - [\[Project Link\]](#)

2024 - 2025

Project Director - Origins of Life Simulation Device

- Led a multidisciplinary capstone team in the design of an automated wet-dry cycling instrument for origins-of-life and RNA/DNA polymerization research, translating biochemistry research needs into engineering requirements and testable system specifications.
- Developed an ESP32-based control system with programmable cycle timing, environmental sensing, safety interlocks, and closed-loop temperature and humidity regulation for repeatable laboratory protocols.
- Designed a modular platform with multiple sample chambers, integrated data logging, remote monitoring, and web-based user interaction to support longer unattended experiments and protocol reconfiguration.
- Managed technical integration, documentation, validation testing, and stakeholder communication with researchers while balancing mechanical design, embedded software, and experimental reliability constraints.

Professor Piano - Wearable Piano Teaching Glove - [\[Project Link\]](#)

March, 2025

Wearable Sensing and Interactive Feedback System

- Designed a wearable piano instruction glove that combined piezoelectric sensors, a 9-axis IMU, and an ESP32 to capture finger force, hand motion, and playing posture in real time.
- Implemented signal-processing pipelines for peak detection, filtering, calibration, and movement interpretation to improve the reliability of finger press and orientation measurements across users.
- Integrated I2S audio playback to provide immediate auditory feedback and support an interactive training experience focused on technique reinforcement and muscle-memory development.
- Contributed to the system concept across electronics, embedded software, and ergonomic wearable design, including flexible integration considerations for comfort, repeatability, and extended use.

Mechatronics Robot Competition - [\[Project Link\]](#)

2024

Team Integration Director - Autonomous Competition Robot

- Directed subsystem integration for a five-week autonomous robot competition project, coordinating mechanical, electrical, and embedded software development under an aggressive build and test schedule.
- Built the robot around a finite state machine architecture with sensor-driven behaviors for navigation, task execution, and fault handling, supporting deterministic operation and rapid iteration during development.
- Designed custom electrical subsystems including analog filters, motor-control electronics, power distribution, and sensor signal-conditioning circuits to improve robustness in a noisy competition environment.
- Managed interface definitions, team timelines, integrated testing, and system troubleshooting as the robot progressed from concept to validated competition platform against 20 teams.

Work Experience

University of California, Santa Cruz

Spring 2026

Incoming Teaching Assistant - Mechatronics

- Selected to support undergraduate mechatronics instruction through hands-on laboratory assignments spanning electrical systems, mechanical integration, embedded software, analog circuits, and motor control.
- Mentor student teams through subsystem debugging, design iteration, and system integration as they build toward the course's final robot competition.
- Translate complex sensing, actuation, control, and fabrication concepts into practical lab guidance that helps students move from theory to a working robotic platform.

Palos Verdes Summer Tech Internship [\[Project Link\]](#)

Summer 2021

Senior Engineering Intern - Canine Gait Analysis Wearable Project

- Directed the development of a wearable diagnostic device intended to help veterinarians detect abnormal gait patterns in dogs earlier and at lower cost than conventional screening workflows.
- Integrated an Arduino Nano, MPU-9250 inertial sensing, battery regulation, MicroSD logging, and harness-mounted packaging into a portable prototype capable of collecting multi-axis motion data during live canine walking trials.
- Conducted data collection and test execution with healthy dog subjects to build baseline datasets for a machine-learning pipeline aimed at distinguishing healthy and abnormal stride behavior.
- Helped connect recorded motion data to downstream visualization and analysis workflows designed to identify likely gait irregularities and improve veterinarian insight.